

W-O-SC-10 Operator Setup Guide

NNBT/Purpald + guaiacol + ABTS + MALDI — laccase activity assay on the OT-2

This guide is a deck-setup reference for the person loading the robot. It does not replace the run log the protocol prints in the Opentrons app at the start of every run – that log has the **exact** volumes, wells and tube contents for the parameters chosen that day. This document explains the fixed structure that never changes: which slot holds what, which tube each reagent goes in, and the order operator actions happen in.

Before you touch the deck

Open the protocol in the Opentrons app and fill in the runtime parameters (number of enzymes, Bradford concentrations, molar masses, tip start positions, MALDI on/off). **Do not edit the .py file.**

The app's Setup screen will show a coloured, named liquid in every tube, reservoir well and plate well once parameters are set – use it to double-check this guide against your specific run before starting.

1. What this run does

One dilution series feeds three readouts from the same enzymes:

- **NNBT plate** (slot 1, on the Heater-Shaker) – laccase activity via NNBT/Purpald, run in two parallel arms: plain and with guaiacol.
- **ABTS plate** (swapped into slot 5) – a fast kinetic activity readout at A414/A734, read immediately after spiking.
- **MALDI target** (swapped into slot 5 later) – optional. Spotted at t = 0, 30, 120 min while the NNBT plate incubates. **One row per sample** (NC2, then the enzymes) with time running across the columns, two columns per timepoint (plain, then guaiacol).
- **10× readout plate** (swapped into slot 5 last) – an empty plate of the **same type as the NNBT plate**. Every finished reaction is diluted 10× into it after the incubation, and it finishes on the Heater-Shaker, whose adapter will not take a Corning flat plate.

The NNBT plate holds up to 10 enzymes plus two negative controls (NC2 = heat-inactivated enzyme, NC3 = active enzyme with no NNBT in the mix) and a 4-point lactaldehyde standard curve, each in triplicate, in both the plain and guaiacol arms.

NC1 (buffer only) is no longer on this plate – ten enzymes need the row, and the 2026-09-15 run showed NC1 and NC2 agreeing to within one plate-noise SD, so NC2 (which is NC1 plus protein) is the better blank and does NC1's job. NC1 still runs on the ABTS plate, out of dilution well B3.

The NNBT plate is read twice. After the incubation, 15 µL of every well goes into 135 µL of milliQ on the readout plate. Both plates then get Purpald and develop on the Heater-Shaker, one after the other, and both are read at A530: save them as **"-undiluted"** and **"-10x"**. The 10× plate is the one whose standards should be on scale – on 2026-09-15 every standard from 500 to 6000 µM read as one flat line on the blank, because the Purpald response peaks below 500 µM and turns brown above it.

2. Deck layout

Slot	Contents	Notes
1	Heater-Shaker + NNBT plate (on the flat-bottom adapter)	Stays on the deck the whole run. North (slot 4) must stay empty.
2, 4, 9, 10, 11	EMPTY	Clearance, not spare space – leave empty.
5	Tube rack → ABTS plate → MALDI target (in that order)	Only one of the three is ever on the deck at a time. The robot pauses and asks you to make each swap.
3	20 µL tip rack	p20 ONLY. Never place a tube rack, reservoir or anything the p300 touches here (see §7).

Slot	Contents	Notes
6	Dilution plate (96-well)	Holds every diluted stock in columns 1–3. Stays on the deck the whole run: the MALDI 1:5 dilutions go in columns 4–12 of the same plate.
7	300 µL tip rack → tube rack (MALDI only) → 300 µL tip rack	If MALDI is on, the robot asks you to take the tip rack out and put the tube rack back here with the capped matrix tube, then to swap them back before Purpald.
8	12-well reagent reservoir	North (slot 11) must stay empty.
12	Fixed trash	Do not move or cover.

Load order for slot 5: put the **tube rack** on before starting the run. Keep the ABTS plate and MALDI target staged off-deck (labelled) until the robot pauses and asks for them by name.

3. Tip racks

Two full racks: a 20 µL rack in slot 3 (p20, right mount) and a 300 µL rack in slot 7 (p300, left mount). In the app, set “20 uL tip row/column” and “300 uL tip column” to the first unused tip if you are reusing partial racks – otherwise leave at A1. The app tells you at run start whether the loaded racks have enough tips; if not, keep a spare full rack of each size on hand, because the run will pause mid-way to ask for one.

4. Tube rack (slot 5, step 1 of 3)

A 24-position 1.5 mL tube rack (4 rows A–D × 6 columns). Enzyme stocks fill column-first: enzyme 1 → A1, enzyme 2 → B1, enzyme 3 → C1, enzyme 4 → D1, enzyme 5 → A2, and so on. Two positions are fixed and never move, regardless of how many enzymes are used:

Tube	Contents	Colour
C6 (fixed)	NC2 – heat-inactivated enzyme stock	
D6 (fixed)	Lactaldehyde stock, 100,000 µM (1 M diluted 1:10)	
A5 (fixed)	PaDa-1 1:1000 – ABTS positive control, made up by hand from powder	
B5, B6	MALDI matrix (DHB/acetonitrile), CAPPED – only if MALDI is on. Added when the rack goes back into slot 7; the second tube only if the run log asks for it.	

Enzyme positions (A1, B1, C1, D1, A2 ...) hold each enzyme's own stock at whatever concentration your Bradford assay measured – the robot dilutes it to the common target concentration itself. The enzyme chosen as “NC3 uses enzyme #” is drawn from its own tube a second time, so nothing is short-pipetted. **Exact tube volumes needed are in the run log** – they depend on how many enzymes you run and the tip use they need to survive.

5. Reagent reservoir (slot 8)

A 12-well reservoir. All reagent mixes below are **premixed by hand off-deck** before the run – the robot only dispenses them. Pour to the volume shown in the run log (it already includes dead volume); the table below shows which well gets which reagent.

Well(s)	Reagent	Colour
A1, A2	NNBT reaction mix (assay buffer + NNBT)	
A3	Assay / dilution buffer	
A4	Purpald reagent	
A5	Laccase ABTS reaction mix	
A6	PaDa-1 ABTS reaction mix – H ₂ O ₂ instead of Cu ²⁺ , and a different pH. Not interchangeable with the laccase mix.	

Well(s)	Reagent	Colour
A7, A8	Guaiacol reaction mix (buffer + NNBT + guaiacol)	
A9	No-NNBT mix (NC3, plain arm)	
A10	No-NNBT + guaiacol mix (NC3, guaiacol arm)	
A11, A12	milliQ water – the 10× readout diluent (always), plus the MALDI 1:5 premix if spotting is on. Both wells: one cannot hold it.	

Wells shown as two positions (e.g. NNBT mix, guaiacol mix) mean the reagent is split across both if a single 15 mL well would not hold enough for a large run – the run log tells you whether the second well is actually needed.

6. Run sequence & where the robot will pause

Steps 1–5, 7–9, 11, 13–14 and 16–17 run automatically. The robot stops and waits for you at every step marked **PAUSE** – read the on-screen message before resuming.

#	Action	Operator note
1	Buffer dispensed into every dilution well	
2	Enzyme, NC3 and heat-inactivated stocks diluted and mixed to target conc.	
3	Lactaldehyde standard curve built by serial dilution	
4	PaDa-1 1:1000 moved from its tube into a dilution well	rack is about to leave
5	10 µL spikes into all four NNBT plate blocks	
6	PAUSE	Swap the tube rack for the ABTS plate in slot 5.
7	10 µL spikes into the ABTS plate, one sample at a time	
8	Reaction mixes dispensed into the NNBT plate	
9	Both ABTS mixes dispensed by column – always last, since ABTS reacts on contact	column 1 takes the PaDa-1 mix, the rest the laccase mix
10	PAUSE	Take the ABTS plate to the reader immediately (read A414/A734), then seal the NNBT plate. If MALDI is on: MALDI target into slot 5, 300 µL tips out of slot 7, tube rack with the capped matrix tube into slot 7.
11	Incubate 2 h @ 40°C	If MALDI is on, pauses at t = 0, 30, 120 min: unseal → open the matrix tube once for the whole session, close it at the end → reseal. Not once per spot.
12	PAUSE for the 10× readout	Remove the plate seal. Put the empty readout plate in slot 5 and a FRESH 20 µL tip rack in slot 3 – the 10× pass spends a whole rack, one tip column per plate column. Check the run log for how many racks this run needs in total.
13	135 µL milliQ into the readout plate, then 15 µL out of every NNBT well into it, mixed	
14	45 µL Purpald into the undiluted plate, developed 10 min	this plate is quenched first on purpose – it still holds active enzyme at full NNBT, while the 10× plate already runs ten times slower
15	PAUSE to swap the plates	Take the undiluted plate off the Heater-Shaker to the reader (A530, save as "-undiluted"), then put the 10× plate onto the Heater-Shaker.
16	10× plate shaken 2 min at 1000 rpm	15 µL under 135 µL does not mix itself, and Purpald meeting a concentrated bolus browns it for good
17	50 µL Purpald into the 10× plate, developed 10 min, done	Read A530, save as "-10x".

7. Three rules that have already cost a run

Rule 1 – Slot 9 can never take a single-nozzle move

The fixed trash in slot 12 sits directly north of slot 9, and the simulator does not model that clearance. Never place labware needing single-tip access in slot 9. The protocol checks this automatically and will refuse to load if violated.

Rule 2 – a HEIGHT is not a DEPTH

A tip positioned above the liquid surface draws air; a tip pressed to a flat well bottom seals against it and moves nothing. Every liquid-handling height in this protocol is computed from the well's remaining volume – if you ever manually jog the robot, do not assume a fixed height works at every fill level.

Rule 3 – the LEFT mount (p300) cannot reach the right-hand deck column

Both pipettes ride the same X carriage ~34 mm apart, against a hard limit at X 418 mm, so the left-mount p300 tops out near deck X 382. Slots 3/6/9 start at x = 265, so anything past x ≈ 117 within one of those slots is physically unreachable by the p300 – and a touch-tip adds another 4 mm on top.

This is what caused the 2026-09-11 run failure: the tube rack was in slot 3 and a touch-tip on tube D6 commanded X422.2, tripping a hard-limit alarm.

Operator rule: slot 3 must only ever hold the 20 µL tip rack. Never place a tube rack, reservoir, or any labware the p300 touches there. The tube rack, ABTS plate and MALDI target all belong in slot 5, which sits in the middle deck column and is reachable by both pipettes. During MALDI only, the tube rack goes in slot 7, where only the p20 visits it.

8. Worked example (3 enzymes) – for illustration only

The tables below are generated from the protocol's own layout code using 3 placeholder enzymes, purely to show the *shape* of the plate maps and what a completed reservoir/tube-rack setup looks like. **Do not pour these volumes for a real run** – use the run log the app prints for your actual parameters.

NNBT plate map (slot 1)

N = NNBT mix · G = guaiacol mix · x = no-NNBT mix (NC3) · y = no-NNBT + guaiacol mix (NC3). Number = group index, keyed to the table below.

	1	2	3	4	5	6	7	8	9	10	11	12
A	N01	N01	N01	G09	G09	G09	N17	N17	N17	G25	G25	G25
B	x02	x02	x02	y10	y10	y10	N18	N18	N18	G26	G26	G26
C	N03	N03	N03	G11	G11	G11	N19	N19	N19	G27	G27	G27
D	N04	N04	N04	G12	G12	G12	N20	N20	N20	G28	G28	G28
E	N05	N05	N05	G13	G13	G13	N21	N21	N21	G29	G29	G29
F	N06	N06	N06	G14	G14	G14	N22	N22	N22	G30	G30	G30
G	N07	N07	N07	G15	G15	G15	N23	N23	N23	G31	G31	G31
H	N08	N08	N08	G16	G16	G16	N24	N24	N24	G32	G32	G32

#	Wells	Content
1	A1,A2,A3	NC2 heat-inactivated
2	B1,B2,B3	NC3 (Lac-01)
3	C1,C2,C3	Lac-std 5000 uM
4	D1,D2,D3	Lac-std 2500 uM
5	E1,E2,E3	Lac-std 1200 uM
6	F1,F2,F3	Lac-std 400 uM

#	Wells	Content
7	G1,G2,G3	(unused row G)
8	H1,H2,H3	(unused row H)
9	A4,A5,A6	NC2 heat-inactivated
10	B4,B5,B6	NC3 (Lac-01)
11	C4,C5,C6	Lac-std 5000 uM
12	D4,D5,D6	Lac-std 2500 uM
13	E4,E5,E6	Lac-std 1200 uM
14	F4,F5,F6	Lac-std 400 uM
15	G4,G5,G6	(unused row G)
16	H4,H5,H6	(unused row H)
17	A7,A8,A9	Lac-01
18	B7,B8,B9	Lac-02
19	C7,C8,C9	Lac-03
20	D7,D8,D9	(unused row D)
21	E7,E8,E9	(unused row E)
22	F7,F8,F9	(unused row F)
23	G7,G8,G9	(unused row G)
24	H7,H8,H9	(unused row H)
25	A10,A11,A12	Lac-01
26	B10,B11,B12	Lac-02
27	C10,C11,C12	Lac-03
28	D10,D11,D12	(unused row D)
29	E10,E11,E12	(unused row E)
30	F10,F11,F12	(unused row F)
31	G10,G11,G12	(unused row G)
32	H10,H11,H12	(unused row H)

ABTS plate map (slot 5, swapped in after the dilutions)

#	Wells	Content	Reaction mix
1	A1,B1,C1	PaDa-1 1:1000	PaDa-1 ABTS mix (H2O2)
2	A2,A3,A4	NC1 buffer only	laccase ABTS mix
3	B2,B3,B4	NC2 heat-inactivated	laccase ABTS mix
4	C2,C3,C4	Lac-01	laccase ABTS mix
5	D2,D3,D4	Lac-02	laccase ABTS mix
6	E2,E3,E4	Lac-03	laccase ABTS mix

The two PaDa-1 strengths are the positive control – there is no ABTS radical standard curve. They run *down* column 1 because the reaction mixes go in by whole columns, and a column can only take one mix. Wells in a part-filled column still receive reaction mix but hold no sample: **they are blank, do not read them.**

Dilution plate (slot 6), 150 µL per well

Well	From	Stock µL	Buffer µL	Content
A1	tube C6	138.8	61.2	NC2 heat-inactivated
B1	tube A1	138.8	61.2	NC3 (Lac-01)
C1	tube D6	150.0	50.0	Lac-std 5000 uM
D1	tube D6	75.0	125.0	Lac-std 2500 uM
E1	tube D6	36.0	164.0	Lac-std 1200 uM
F1	tube D6	12.0	188.0	Lac-std 400 uM
G1	res A3	0.0	200.0	(unused row G)
H1	res A3	0.0	200.0	(unused row H)
A2	tube A1	138.8	61.2	Lac-01
B2	tube B1	180.0	20.0	Lac-02
C2	tube C1	115.7	84.3	Lac-03
D2	res A3	0.0	200.0	(unused row D)
E2	res A3	0.0	200.0	(unused row E)
F2	res A3	0.0	200.0	(unused row F)
G2	res A3	0.0	200.0	(unused row G)
H2	res A3	0.0	200.0	(unused row H)
A3	tube A5	0.0	0.0	PaDa-1 1:1000
B3	res A3	0.0	100.0	NC1 buffer only

Reservoir (slot 8) – example pour volumes

Well	Reagent	Volume
A1	NNBT mix	8.2 mL
A3	assay buffer	3.3 mL
A4	Purpald reagent	11.2 mL
A5	laccase ABTS mix	5.6 mL
A6	PaDa-1 ABTS mix (H2O2)	2.5 mL

Well	Reagent	Volume
A7	guaiacol mix	8.2 mL
A9	no-NNBT mix	1.5 mL
A10	no-NNBT guaiacol mix	1.5 mL

Tube rack – example volumes needed

Tube	Contents	Min. volume
A1	NC3 (Lac-01) + Lac-01	≥ 578 µL
B1	Lac-02	≥ 480 µL
C1	Lac-03	≥ 416 µL
A5	PaDa-1 1:1000	≥ 340 µL
C6	NC2 heat-inactivated	≥ 439 µL
D6	Lac-std 5000 uM + Lac-std 2500 uM + Lac-std 1200 uM + Lac-std 400 uM	≥ 573 µL

Generated from Lac_W-O-SC-10_nnbt_guaiacol_abts_maldi_ot2.py – Kristine Toft Johansen (s215098). This guide reflects the protocol as of the "slot-swap left-mount reach" fix (tube rack confined to slot 5; slot 3 reserved for the 20 µL tip rack only).